

informed consent is most difficult.

While implants could make lives easier, more comfortable and efficient, it would simultaneously become very easy for governments and surveillance agencies to continuously track us.

Microchips may reduce freedom of choice. Implanting microchips may become mandatory for citizens for, say, to apply for a job, getting insurance, be admitted to school, get due benefits and concessions from governments, etc. This could mean that citizens are compelled to pay with microchip implants, too.

This raises certain questions. Will it become mandatory to link the microchip implant for opening a bank account or buying a SIM card for a phone? How will people remove the implant? To provide improved and better public service, service provider(s) may insist on all their employees getting microchip implants. And all these aspects can severely limit the freedom of choice.

Chips could get hacked or corrupted. Since microchip implants contain considerable amount of important personal information, these can become the prime target for people with criminal intentions, such as hackers, kidnappers and the like. Imagine the information on the chip being not only readable but also writable. That would mean data could be corrupted, wiped or copied. Criminals could use the data and copy or replace it with their own data, altering your—and their—physical identity.

As is the case with most new technological developments, implanted microchips are also sensitive to exploitation.

Closing thoughts

The future of micro-chipping in humans is exciting, with the likelihood



Fig. 9: One day we can expect to see brain implants for perfect memory, superior vision, hyper-normal focus and more (Credit: www.wsj.com)

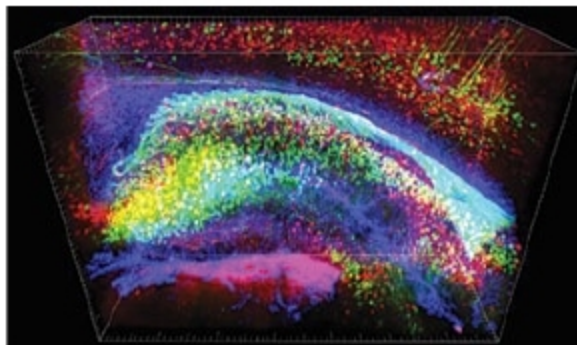


Fig. 10: Research is in progress to test a brain implant device to ultimately cure Alzheimer's disease (Credit: <http://edition.cnn.com>)

of many interesting potential applications becoming commonplace in the near future. Applications include animal tracking, product tracking, inventory control, access and passport control, and so on. New developments in technology have opened up new expectations in the fabrication of tiny devices, which could be implanted in the human body for locating and tracking persons, or for remotely controlling human biological functions.

A microchip implanted inside the human body can transmit its data to an off-body interrogator at a short distance (typically 10cm or less). It cannot communicate with GPS satellites or other equivalent satellite-based positioning systems. However, a remote and reversible control of human bio-molecular machinery is now within reach using existing technology.

While the thought of having a chip injected into the body is a little strange, it really is no stranger than other implanted devices used today, such as pacemakers. Today, micro-chipping is purely voluntary and mainly used for high-risk patients, such as diabetics and people suffering from Alzheimer's and cardiovascular diseases. It may soon become necessary to implant more than just one microchip to accommodate many different digital identification systems and different cards being used.

If human micro-chipping is not done in a mandatory manner, and not used for tracking or frisking, then people generally seem quite open to potential applications. Mass implantation of these chips will, however, need to undergo many debates about health and privacy issues. Points such as how the technology is being utilised by humans, where it is being applied and for what purpose require a commensurate ethical response from the broader community.

For instance, what does it mean for technology to be embedded beneath the skin in a perfectly-healthy human being for the purposes of easy identification or even amplification? Is an implant for non-medical purposes a basic breach of human rights? Are implant IDs, even if consented, in direct conflict with a state's privacy laws? And what happens if an implant cannot be removed on demand because it has become intertwined with tissues in the body? Will we still have control over our body's privacy? Will we be able to remove the implant(s) ourselves? How will we know if our chips are being hacked? Hence, several questions need to be answered in the journey, before we see high proliferation of implants in our day-to-day life. **EFY**

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